

Decision Tree is a Supervised Machine Learning algorithm used for classification & regression

- It works by splitting data into smaller subsets based on different conditions
- The structure looks like an inverted tree, starting from a root node & splitting into branches & leaves.

Concept

- Each internal node represents a test on an attribute (a question)
- Each branch represents the outcome of the test.
- Each leaf node represents a final decision or class label.

Eg. - Question in a decision tree:
"Is the temperature hot, mild, or cool?"

Depending on the answer, the tree moves to the next question until it reaches a decision (like play, don't play).

Numerical Decision Tree (ID3 Algorithm)

	Pay	Outlook	Temp	Humidity	Wind	Target
						Play/Tennis
D1		Sunny	Hot	High	Weak	No.
D2		Sunny	Hot	High	Strong	No
D3		Overcast	Hot	High	Weak	Yes
D4		Rain	Mild	High	Weak	Yes
D5		Rain	Cool	Normal	Weak	Yes
D6		Rain	Cool	Normal	Strong	No
D7		Overcast	Cool	Normal	Strong	Yes
D8		Sunny	Mild	High	Weak	No
D9		Sunny	Cool	Normal	Weak	Yes
D10		Rain	Mild	Normal	Weak	Yes
D11		Sunny	Mild	Abnormal	Strong	Yes
D12		Overcast	Mild	High	Strong	Yes
D13		Overcast	Hot	Normal	Weak	Yes
D14		Rain	Mild	High	Strong	No

$N = 14$ (No of data)

Calculate Entropy of whole dataset, $\text{play}_{\text{tennis}}$ $n=2$

Entropy (S) = $-\sum_{i=1}^n p_i \log_2(p_i)$

$$H(S) = - \left(\frac{9}{14} \log_2 \left(\frac{9}{14} \right) + \frac{5}{14} \log_2 \left(\frac{5}{14} \right) \right)$$

$$= 0.94$$

Now 2) Calculate Entropy for each feature (attributing just node will have highest gain)

Attribute: Outlook

Values (Outlook) = Sunny, Overcast, Rain

Entropy (S_{sunny}) =

Play

Outlook:

Sunny $\rightarrow$ 5

Yes No
2 3

Overcast $\rightarrow$ 4

4 0

Rain $\rightarrow$ 5

3 2

$$\text{Entropy}(S_{\text{sunny}}) = - \left(\frac{2}{5} \log_2 \left(\frac{2}{5} \right) + \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right)$$

$H(S_{\text{sunny}})$

$$= 0.971$$

0

$$\text{Entropy}(S_{\text{overcast}}) = - \left(\frac{4}{4} \log_2 \left(\frac{4}{4} \right) + \frac{0}{4} \log_2 \left(\frac{0}{4} \right) \right)$$

$$= 0$$

pure

$$\text{Entropy}(S_{\text{rain}}) = - \left(\frac{3}{5} \log_2 \left(\frac{3}{5} \right) + \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right)$$

$$= 0.971$$

Now 19 ($S_{\text{attribute}}$)

Gain ($S_{\text{attribute}}$) =

$$= \text{Entropy}(S) - \sum \left(\frac{S_i}{S} \right) \text{Entropy}(S_i)$$

where S is total data
and S_i is data for each attribute

$$= \text{Entropy}(S) - \left(\frac{5}{14} \text{Entropy}(S_{\text{sunny}}) + \frac{4}{14} \text{Entropy}(S_{\text{overcast}}) + \frac{5}{14} \text{Entropy}(S_{\text{rain}}) \right)$$

$$= 0.94 - \left(\frac{5}{14} * 0.971 + \frac{4}{14} * 0 + \frac{5}{14} * 0.971 \right)$$

$$= 0.246$$

$$\text{Gain}(S_{\text{outlook}}) = 0.94 - 0.694$$

$$= 0.246$$

Attribute: Temp

Value (Temp) = Hot, Mild, Cool ✓

Temp: ✓

hot →	4	yes	no
mild →	6	4	2
cool →	4	3	1

$$\text{Entropy}(S_{\text{hot}}) = - \left(\frac{2}{4} \log_2 \left(\frac{2}{4} \right) + \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \right)$$

= 1 (Directly, when classes are equal)

$$\text{Entropy}(S_{\text{mild}}) = - \left(\frac{4}{6} \log_2 \left(\frac{4}{6} \right) + \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \right)$$

$$= 0.918$$

$$\text{Entropy}(S_{\text{cool}}) = - \left(\frac{3}{4} \log_2 \left(\frac{3}{4} \right) + \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \right)$$

$$= 0.811$$

1. Information Gain

Entropy(S)

Date _____
Page _____

$$\text{Gain}(S, \text{Temp}) = H(S) - \sum_{v \in \{\text{hot, mild, cool}\}} \frac{|S_v|}{|S|} H(S_v)$$

$$= 0.94 - \left(\frac{4}{14} * 1 + \frac{6}{14} * 0.918 + \frac{4}{14} * 0.811 \right)$$

$$= 0.94 - 0.911$$

$$= 0.029$$

Again

Attribute: humidity

Value (humidity): ~~weak, strong~~
high, normal

humidity	high	normal	Yes	No
	7	6	3	4

$$\text{Entropy}(S_{\text{humidity}}) = - \left(\frac{3}{7} \log_2 \left(\frac{3}{7} \right) + \frac{4}{7} \log_2 \left(\frac{4}{7} \right) \right)$$

$$= 0.985$$

$$\text{Entropy}(S_{\text{normal}}) = - \left(\frac{6}{7} \log_2 \left(\frac{6}{7} \right) + \frac{1}{7} \log_2 \left(\frac{1}{7} \right) \right)$$

$$\text{Now,} \quad = 0.592$$

$$\text{Gain}(S, \text{humidity}) = H(S) - \sum_{v \in \{\text{high, normal}\}} \frac{|S_v|}{|S|} H(S_v)$$

$$= 0.94 - \left(\frac{7}{14} \times 0.985 + \frac{7}{14} \times 0.592 \right)$$

$$= 0.152$$

Attribute: wind

value (wind): weak, strong

wind	weak	strong	Yes	No
	8	6	6	2

$$\text{Entropy}(S_{\text{weak}}) = - \left(\frac{6}{8} \log_2 \left(\frac{6}{8} \right) + \frac{2}{8} \log_2 \left(\frac{2}{8} \right) \right)$$

$$= 0.811$$

$$\text{Entropy}(S_{\text{strong}}) = - \left(\frac{8}{6} \log_2 \left(\frac{8}{6} \right) + \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \right)$$

$$= 1 \quad (\text{Directly})$$

(both equal)

Now,

$$\text{Gain}(S, \text{wind}) = H(S) - \sum_{v \in \{\text{weak, strong}\}} \frac{|S_v|}{|S|} H(S_v)$$

$$= 0.94 - \left(\frac{8}{14} \times 0.811 + \frac{6}{14} \times 1 \right)$$

$$= 0.24 - 0.906$$

$$= -0.035$$

Highest Information Gain is,

of outlook $\rightarrow 0.2464$

So we consider Outlook a root node.

$S \rightarrow D1, D2, \dots, D14 \Rightarrow$ Take

$[9 \rightarrow \text{Yes}] \quad 14$

$5 \rightarrow \text{No}$

Outlook



Sunny

$[D1, D2, D3, D9, D11]$

5

$2 \rightarrow \text{Yes}$

$3 \rightarrow \text{No}$

Overcast

$[D3, D9, D12, D13]$

4

$4 \rightarrow \text{Yes}$

$0 \rightarrow \text{No}$

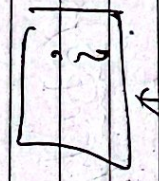
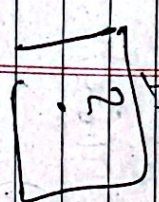
Rain

$[D4, D5, D6, D7, D8, D10]$

5

$3 \rightarrow \text{Yes}$

$2 \rightarrow \text{No}$



For sunny only dataset.

Recursively Repeat the process

Day	Temp	Humidity	Wind	Play
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

(Remember, there is no outlook column here)

Entropy of this dataset,

$$\text{Entropy}(S_{\text{sunny}}) = - \left(\frac{2}{5} \log_2 \left(\frac{2}{5} \right) + \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right)$$

$$= 0.971$$

Now Attribute: - Temp

Values $\rightarrow$ Hot, mild, cool

Temp	Hot	Mild	Cool	Yes	No
	2	1	1	0	2
				1	0

$$\text{Entropy}(S_{\text{Hot}}) = \text{Entropy}(0) \text{ (one sided)}$$

$$\text{Entropy}(S_{\text{Mild}}) = \text{Entropy}(1) \text{ (Both equal)}$$

$$\text{Entropy}(S_{\text{Cool}}) = \text{Entropy}(0) \text{ (one sided)}$$

$$Gain(Sunny, Temp) = H(S) - \sum_{\text{value in } S} \frac{|S_v|}{|S|} H(S_v)$$

$$= 0.921 - \left(\frac{2}{5} * 0 + \frac{2 \times 1}{5} \right) + \frac{1}{5} * 0$$

$$= 0.591$$

Again,

for Attribute Humidity,

values(humidity) $\rightarrow$ High, Normal

humidity $\rightarrow$ High	3	0	3
Normal	2	2	0
		Yes	No

$$\text{Entropy (High)} = \text{Directly } 0 \text{ (one sided)}$$

$$\text{Entropy (Normal)} = \text{Directly } 1 \text{ (one sided)}$$

$$Gain(Sunny, Humidity)$$

$$= H(S_{\text{sunny}}) - \sum_{\text{value in } S} \frac{|S_v|}{|S|} H(S_v)$$

$$= 0.921 - \left(\frac{3}{5} * 0 + \frac{2 \times 0}{5} \right)$$

$$= 0.921$$

Again, for Attribute wind,

values (wind) $\rightarrow$ weak, strong

wind $\rightarrow$ weak	3	1	2
strong	2	1	1
		Yes	No

$$\text{Entropy (weak)} = - \left(\frac{1}{5} \log_2 \left(\frac{1}{3} \right) + \frac{2}{5} \log_2 \left(\frac{2}{3} \right) \right)$$

$$= 0.918$$

$$\text{Entropy (strong)} = \text{directly } 1 \text{ (both equal)}$$

$$Gain(S_{Sunny}, wind) = W(S_{Sunny}) - \sum \frac{|S_v|}{|S|} \times W(S_v)$$

$$= 0.921 - \left(\frac{3}{5} \times 0.928 + \frac{2}{5} \times 1 \right)$$

$$= 0.921 - 0.951$$

$$= -0.030$$

Answer

$$Gain(S_{Sunny}, Temp) = 0.570$$

Ans.

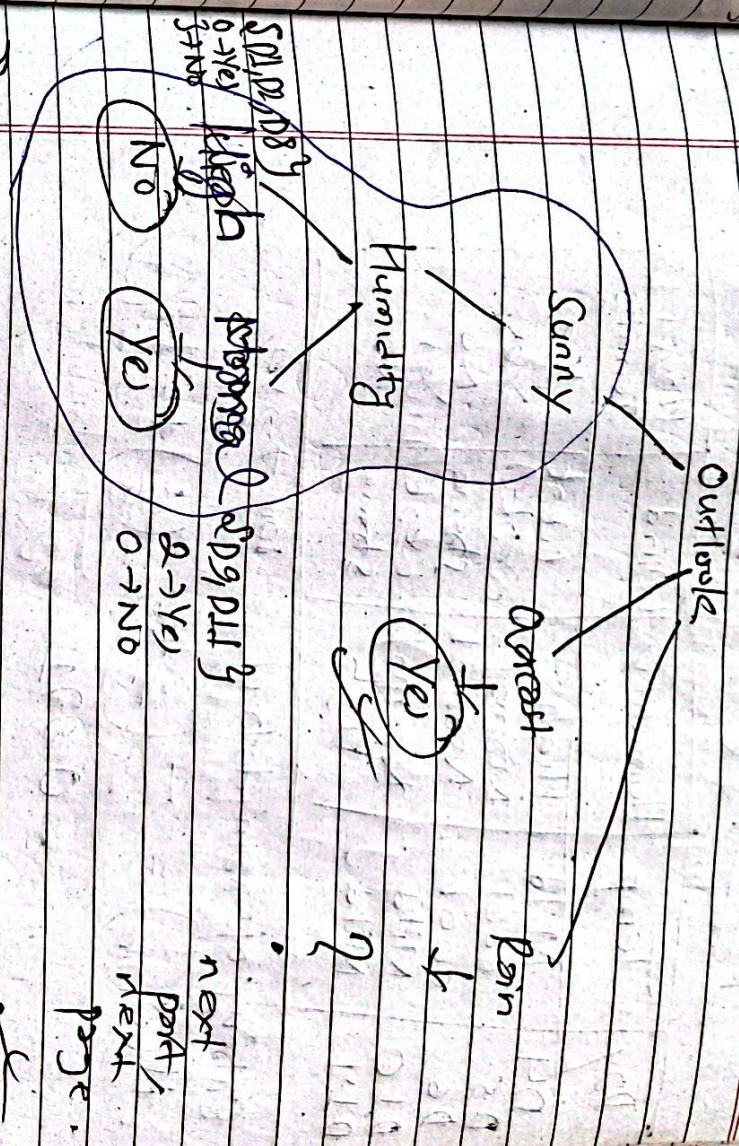
$$Gain(S_{Sunny}, Humidity) = 0.97$$

Highest Gain
So, next node.

$$Gain(S_{Sunny}, Wind) = 0.022$$

Humidity	Play tennis
High	No
High	No
High	No
Normal	Yes
Normal	Yes

Correctly
(covered)



New? for Rain Node

Rain only dataset

Pay	Temp	Humidity	Wind	Play
D4	Mild	High	weak	Yes
D5	cool	Normal	weak	No
D6	cool	Normal	strong	Yes
D10	Mild	Normal	weak	No
D14	Mild	High	strong	No

Entropy of rain dataset 2 no

$$H(S_{rain}) = - \left(\frac{3}{5} \log_2 \left(\frac{3}{5} \right) + \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right) = 0.921$$

Again,

for attribute temp,

values (temp) = mild, cool, hot

temp	hot	0	Yes	No
	mild	3	2	1
	cool	2	1	1

So, Entropy (S_{temp}) = 0

$$\text{Entropy}(S_{mild}) = - \left(\frac{2}{3} \log_2 \left(\frac{2}{3} \right) + \frac{4}{3} \log_2 \left(\frac{1}{3} \right) \right)$$

$$= 0.918$$

Again, Entropy (S_{cool}) = 1 directly (both equal)

Again, Entropy (S_{temp}) = $H(S_{rain}) - \sum \frac{|S_{temp}|}{|S_{rain}|} H(S_{temp})$

$$= 0.921 - \left(0 \times 0 + \frac{3}{5} + 0.918 + \frac{2}{5} \times 1 \right) = 0.020$$

Again, for attribute Humidity, values (Humidity) = Normal, High

Humidity	Normal	3	2	Yes	No
	high	2	1	1	1

$$S_0, \text{Entropy}(S_{\text{normal}}) = - \left(\frac{2}{3} \log_2 \left(\frac{2}{3} \right) + \frac{1}{3} \log_2 \left(\frac{1}{3} \right) \right)$$

$$= 0.918$$

$$\text{Entropy}(S_{\text{rich}}) = 1 \text{ Directly (both equal)}$$

$$\text{Abol/} \text{gan}(S_{\text{rain, humidity}}) =$$

$$H(S_{\text{rain}}) = - \sum_{v \in \{\text{normal, high}\}} \frac{1}{5} \log_2 \left(\frac{1}{5} \right)$$

$$= 0.921 - \left(\frac{3}{5} \log_2 \left(\frac{3}{5} \right) + \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right)$$

$$= 0.920$$

Again,

for

Absolute wind,

Values (wind) $\rightarrow$ weak, strong

wind $\rightarrow$ weak 3 3 0
strong 2 0 2

$$\text{Entropy}(S_{\text{weak}}) = - \left(\frac{3}{6} \log_2 \left(\frac{3}{6} \right) + \frac{3}{6} \log_2 \left(\frac{3}{6} \right) + \frac{0}{6} \log_2 \left(\frac{0}{6} \right) \right) \text{ directly (one sided)}$$

$$\text{Entropy}(S_{\text{strong}}) = 0 \text{ directly (one sided)}$$

NO,

$$\text{gan}(S_{\text{rain, wind}}) = H(S_{\text{rain}}) = - \sum_{v \in \{\text{normal, high}\}} \frac{1}{5} \log_2 \left(\frac{1}{5} \right)$$

$$= H(S_{\text{rain}})$$

$$= 0.921$$

$$\text{So, } \text{gan}(S_{\text{rain, Temp}}) = 0.920$$

$$\text{gan}(S_{\text{rain, humidity}}) = 0.920$$

$$\text{gan}(S_{\text{rain, wind}}) = 0.921 \text{ Highest so our next node}$$

SP1, P2, ... P14

Yes → 9
No → 5

outlook

Sunny

Overcast

Rainy

Humidity

Yes

Wind

High

Normal

Weak

Strong

No

Yes

Yes

No

SP1, SP2, SP3, SP4, SP5, SP6, SP7, SP8, SP9, SP10, SP11, SP12, SP13, SP14

SP4, SP5, SP6, SP16, P17

✓